

## CAPE PEEL WEST, NU, Canada

WMO: 710640

Lat: 69.036N

Lon: 107.823W

Elev: 542

StdP: 14.41

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-----|-----|------|-----------------------|-------|------|-------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99% |     |      | 0.4%                  |       | 1%   |       |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB  | WS   | MCDB  | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)   | (l)  | (m)   | (n)                      | (o)  | (p)   |
| 2             | -40.6      | -37.6 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 43.4                  | -16.2 | 39.9 | -17.8 | 13.2                     | 340  | 1.160 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 7             | 12.4                   | 66.1            | N/A  | 62.0 | N/A  | 58.3 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 12.2                 | 230  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 39.1          | 34.3 | 29.9 |                            |       |                    |     |                                                    |       |            |       |            |       |            |       |      |     |
| (5)               |      |      | DB                         | -43.5 | 70.4               | 3.8 | 4.4                                                | -46.2 | 73.6       | -48.5 | 76.2       | -50.6 | 78.7       | -53.4 | 81.9 |     |
|                   |      |      | WB                         | N/A   | N/A                | N/A | N/A                                                | N/A   | N/A        | N/A   | N/A        | N/A   | N/A        | N/A   | N/A  |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.2    | -22.3 | -22.5 | -16.9 | -3.5  | 16.0  | 37.5  | 49.6  | 45.0  | 32.1  | 13.9  | -4.6  | -16.7 |
|                                                                        | DBStd       | 27.35  | 9.37  | 8.74  | 11.06 | 9.81  | 9.33  | 8.60  | 6.12  | 6.87  | 7.06  | 9.83  | 10.02 | 9.59  |
|                                                                        | HDD50       | 15022  | 2240  | 2030  | 2072  | 1604  | 1053  | 387   | 84    | 187   | 537   | 1120  | 1639  | 2069  |
|                                                                        | HDD65       | 20383  | 2705  | 2450  | 2537  | 2054  | 1518  | 825   | 479   | 620   | 987   | 1585  | 2089  | 2534  |
|                                                                        | CDD50       | 116    | 0     | 0     | 0     | 0     | 0     | 12    | 71    | 33    | 0     | 0     | 0     | 0     |
|                                                                        | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 14.8   | 15.2  | 15.2  | 15.1  | 15.4  | 15.0  | 13.8  | 13.2  | 12.9  | 15.4  | 15.8  | 14.8  | 15.6  |
|                                                                        | PrecAvg     | 7.0    | 0.2   | 0.2   | 0.3   | 0.3   | 0.3   | 0.7   | 1.3   | 1.4   | 0.9   | 0.7   | 0.4   | 0.3   |
| (15) Precipitation                                                     | PrecMax     | 9.2    | 0.5   | 0.5   | 1.1   | 0.9   | 0.9   | 1.7   | 2.5   | 2.8   | 2.2   | 1.3   | 0.8   | 0.6   |
|                                                                        | PrecMin     | 4.7    | 0.0   | 0.0   | 0.1   | 0.1   | 0.0   | 0.1   | 0.4   | 0.7   | 0.4   | 0.3   | 0.2   | 0.1   |
|                                                                        | PrecStd     | 1.1    | 0.1   | 0.1   | 0.2   | 0.2   | 0.2   | 0.4   | 0.6   | 0.6   | 0.4   | 0.3   | 0.2   | 0.1   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 9.9   | 6.5   | 16.2  | 25.0  | 36.7  | 66.5  | 72.5  | 70.9  | 51.4  | 34.2  | 23.6  | 15.9  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | DB     | -0.2  | 0.4   | 9.7   | 18.8  | 33.7  | 60.1  | 67.5  | 64.5  | 47.3  | 32.1  | 18.7  | 8.2   |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | DB     | -4.4  | -4.4  | 5.1   | 14.4  | 31.5  | 55.5  | 63.6  | 59.7  | 44.5  | 30.1  | 15.0  | 3.1   |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | DB     | -8.4  | -9.4  | -0.1  | 10.3  | 29.0  | 50.5  | 60.2  | 55.5  | 41.7  | 27.5  | 10.9  | -2.5  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5% DB       | MDBR   | 9.8   | 10.7  | 11.2  | 12.4  | 10.1  | 10.0  | 12.4  | 10.8  | 7.6   | 9.5   | 10.6  | 10.1  |
|                                                                        |             | MCDBR  | 14.6  | 16.1  | 13.6  | 14.5  | 9.7   | 17.3  | 17.7  | 16.8  | 11.7  | 8.5   | 12.7  | 15.3  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDWR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5% WB       | MCDWR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.144 | 0.206 | 0.237 | 0.286 | 0.313 | 0.345 | 0.350 | 0.325 | 0.286 | 0.228 | 0.157 | N/A   |
|                                                                        | taud        |        | 1.901 | 2.059 | 2.237 | 2.138 | 2.139 | 2.275 | 2.378 | 2.483 | 2.497 | 2.161 | 1.891 | N/A   |
|                                                                        | Ebn at Noon |        | 27    | 185   | 258   | 271   | 275   | 269   | 264   | 259   | 241   | 170   | 16    | 0     |
|                                                                        | Edh at Noon |        | 1     | 13    | 22    | 34    | 38    | 35    | 31    | 25    | 18    | 12    | 1     | 0     |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 6     | 136   | 634   | 1394  | 2003  | 1915  | 1673  | 1083  | 520   | 168   | 20    | 0     |
|                                                                        | RadStd      |        | 1     | 12    | 38    | 71    | 105   | 99    | 141   | 89    | 55    | 25    | 2     | 0     |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page